

TWO EQUIVALENT PARAMETRIZATIONS OF COHOMOLOGICAL REPRESENTATIONS OF THE UNITARY GROUP $U(p, q)$

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ABSTRACT. In this paper we explain the two different parameterizations of cohomological representations of $U(p, q)$ which had been developed by Mathilde-Gerbelli Gauthier and Bergeron; and provide a concrete bijection between the two parameterizations.

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1. INTRODUCTION

Let G be a semisimple Lie group with maximal compact subgroup K . Matsushima's formula [Mat67] describes how the cohomology of any cocompact lattice $\Gamma \subset G$ can be decomposed as

$$H^i(\Gamma, \mathbb{C}) \simeq H_{dR}^i(\Gamma \backslash G/K, \mathbb{C}) = \bigoplus_{\pi \in \hat{G}} m(\pi, \Gamma) H^i(\mathfrak{g}, K; \pi_\infty).$$

Here $\hat{G}$ is the unitary dual of G , i.e. the set of all unitary irreducible representations of G up to unitary isomorphism, the integer $m(\pi, \Gamma)$ is the multiplicity of π in the decomposition of $L^2(\Gamma \backslash G)$ into irreducibles, and $H^i(\mathfrak{g}, K; \pi_\infty)$ is the relative Lie algebra cohomology of the $(\mathfrak{g}, K)$ -module π_∞ of smooth, K -finite vectors in π . In view of this decomposition, it is important to understand cohomological $(\mathfrak{g}, K)$ -modules, i.e. the $(\mathfrak{g}, K)$ -modules V such that

$$H^i(\mathfrak{g}, K; V) \neq 0.$$

In [VZ84], Vogan and Zuckerman construct and classify the cohomological $(\mathfrak{g}, K)$ -modules of G . They attach a representation $A_{\mathfrak{q}}$ to each K -conjugacy class of so-called θ -stable parabolic subalgebra $\mathfrak{q}$ of the Lie algebra G and show that these $A_{\mathfrak{q}}$ exhaust all possible cohomological representations. This reduces the problem to understanding the isomorphism classes of $\mathfrak{q}$, and when they give rise to isomorphic $A_{\mathfrak{q}}$.

In [DGG22], Dalal–Gerbelli–Gauthier focus on the unitary group $G = U(p, q)$. They give concrete computations of the cohomological representations of G and parametrize them via objects called bipartitions. Also studying cohomological representations of unitary groups, but from a different perspective, Bergeron [Ber09] parametrizes cohomological representations by associating to each $A_{\mathfrak{q}}$ a pair of Young diagrams (λ_+, λ_-) satisfying certain conditions.

Each of these two parameterizations encodes some information about the representations: the construction of Dalal–Gerbelli–Gauthier contains information about where the $A_{\mathfrak{q}}$ sit in the Langlands classification, and specifically, it remembers the cohomological Arthur packets to which a given $A_{\mathfrak{q}}$ belongs, see [MR19]. On the other hand, Bergeron's Young tableaux encode information about the geometry of the Shimura variety $\Gamma \backslash G/K$, specifically about the behaviour of the cohomology classes upon restriction to subvarieties, or under the cup product operation [BMM16].

It is therefore desirable to show the equivalence of these two parameterizations and to give a translation between them. This is what we do in this paper.

2. THETA-STABLE SUBALGEBRAS: DEFINITIONS AND CONSTRUCTIONS

2.1. General construction. As described in the introduction, Vogan-Zuckerman [VZ84] show that every cohomological representation of G is constructed from a θ -stable parabolic subalgebra $\mathfrak{q}$. In what follows, we will follow their convention: given a Lie group G , we will denote its real Lie algebra by $\mathfrak{g}_0$ and its complexification by $\mathfrak{g}$. Let $\mathfrak{g}$ be the complexified Lie algebra of G , and let $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ be a Cartan decomposition, with corresponding Cartan involution θ . Let $x \in i\mathfrak{k}_0$ and let $\text{ad}(x) : \mathfrak{g} \rightarrow \mathfrak{g}$ be the action of x by the adjoint representation. Then the eigenvalues of $\text{ad}(x)$ are real by construction. The θ -stable parabolic subalgebra $\mathfrak{q} = \mathfrak{q}(x)$ attached to x is then defined as

$$\mathfrak{q} = \mathfrak{l} \oplus \mathfrak{u}$$

where $\mathfrak{l}$ is the sum of the eigenspaces of $\text{ad}(x)$ with eigenvalue 0, and $\mathfrak{u}$ is the sum of the eigenspaces of $\text{ad}(x)$ with positive eigenvalue. To each such θ -stable parabolic subalgebra $\mathfrak{q}$, Vogan-Zuckerman attach a $(\mathfrak{g}, K)$ -module $A_{\mathfrak{q}}$, constructed via cohomological parabolic induction, and they show that every $(\mathfrak{g}, K)$ -module V such that

$$H^*(\mathfrak{g}, K; V) \neq 0$$

is of the form $A_{\mathfrak{q}}$ for some $\mathfrak{q}$. We next move to an explicit description of the $\mathfrak{q}$ for the group $G = U(p, q)$.

2.2. Unitary Groups. Let

$$G = U(p, q) = \{g \in GL_{p+q}(\mathbb{C}) : g\Phi_{p+q}\bar{g}^t = \Phi_{p+q}\}, \quad \Phi_{p+q} = \begin{pmatrix} I_p & \\ & -I_q \end{pmatrix}.$$

Then

$$K = \left\{ \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} \in G : A \in U(p), D \in U(q) \right\}$$

defines a choice of maximal compact subgroup with Cartan subgroup

$$T = \left\{ g \in K : g = \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} \text{ with } A, D \text{ diagonal} \right\}.$$

Note that the entries of A, D in the above are complex numbers of norm 1.

We have $\mathfrak{g} \simeq M_{(p+q) \times (p+q)}(\mathbb{C})$ and $\mathfrak{k} \simeq M_{p \times p}(\mathbb{C}) \times M_{q \times q}(\mathbb{C})$ embedded block-diagonally. The corresponding Cartan involution is given by

$$\theta(g) = \Phi_{p+q} g \Phi_{p+q}^{-1},$$

giving rise to the decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ where $\mathfrak{p}$ consists of block off-diagonal matrices.

We identify $\mathfrak{t}$ with the diagonal matrices in $\mathfrak{g}$, and denote its elements by $\{x_1, \dots, x_p, y_1, \dots, y_q\}$. Note that $\mathfrak{t}_0 \simeq (i\mathbb{R}, \dots, i\mathbb{R})$ so that the entries of $X \in i\mathfrak{t}_0$ are real.

3. DALAL–GERBELLI–GAUTHIER'S PARAMETRIZATION BY BIPARTITIONS

3.1. Setup. We begin by introducing the notion of bipartition of a pair (p, q) of natural numbers.

Definition 3.1. An *ordered bipartition* of (p, q) is an ordered tuple of pairs $((p_1, q_1), \dots, (p_r, q_r))$ where $p, q, r \in \mathbb{Z}^{\geq 0}$ with $p_i + q_i > 0$ and $\sum_i p_i = p$, $\sum_i q_i = q$. Let $\mathcal{P}(p, q)$ denote the set of all ordered bipartitions of (p, q) .

Definition 3.2. Define $\mathcal{P}_1(p, q)$ to be the subset of $\mathcal{P}(p, q)$ consisting of ordered tuples $((p_1, q_1), \dots, (p_r, q_r))$ where for each i , if either of p_i or q_i is 0, then the other is 1. For example, $((1, 2), (0, 2), (5, 1)) \notin \mathcal{P}_1(6, 5)$ but $((1, 2), (0, 1), (0, 1), (5, 1)) \in \mathcal{P}_1(6, 5)$.

Definition 3.3. An ordered bipartition $((p_1, q_1), (p_2, q_2), \dots, (p_r, q_r))$ is said to be *maximally compact* if it does not contain two consecutive pairs of the forms $(n, 0), (m, 0)$ and $(0, n), (0, m)$. Let $\mathcal{P}_c(p, q)$ denote the set of maximally compact ordered bipartitions. For example, $((2, 0), (3, 0), (1, 2)) \notin \mathcal{P}_c(6, 2)$ but $((5, 0), (1, 2)) \in \mathcal{P}_c(6, 2)$.

3.2. Parametrization. Let $\gamma : \mathcal{P}(p, q) \rightarrow \mathcal{P}_1(p, q)$ be the natural surjective map where γ replaces any term of the form $(n, 0)$ (and respectively $(0, n)$) with n copies of $(1, 0)$ (or $(0, 1)$). Proposition 11.1.2 of [DGG22], which summarizes results of [Tra01], shows that the set of all θ -stable parabolic subalgebras of $U(p, q)$ is in bijection with the set $\mathcal{P}(p, q)$ and that the cohomological representations of $U(p, q)$ are in bijection with the set $\mathcal{P}_1(p, q)$. We note for further use that instead of $\mathcal{P}_1(p, q)$, we could have used $\mathcal{P}_c(p, q)$ as a result of the following lemma.

Lemma 3.4. *There is a bijection between $\mathcal{P}_1(p, q)$ and $\mathcal{P}_c(p, q)$.*

Proof. The restriction of γ to the subset $\mathcal{P}_c(p, q)$ is the desired bijective map from $\mathcal{P}_c(p, q)$ to $\mathcal{P}_1(p, q)$. The preimage of any $P \in \mathcal{P}_1(p, q)$ contains a unique bipartition B in which all of the consecutive instances of $(1, 0)$ or $(0, 1)$ have been collapsed into a single term, so that $B \in \mathcal{P}_c(p, q)$. $\square$

4. BERGERON'S PARAMETRIZATION BY PAIRS OF YOUNG DIAGRAMS

4.1. Setup. Since we are only considering θ -stable parabolic subalgebras up to conjugation by K , we can first conjugate our element $X \in i\mathfrak{k}_0$ so that $X \in i\mathfrak{t}_0$. Moreover, we can fix a Borel subalgebra $\mathfrak{b}$ of $\mathfrak{k}$ containing $\mathfrak{t}$, and for each equivalence class of $\mathfrak{q}$, choose a representative such that $\mathfrak{q} \cap \mathfrak{k} \supset \mathfrak{b}$. We make such a choice, and specifically choose $\mathfrak{b}$ such that each of our representatives will be of the form $\mathfrak{q} = \mathfrak{q}(X)$ for $X = (x_1, \dots, x_p, y_1, \dots, y_q) \in i\mathfrak{t}_0$ with

$$(1) \quad x_1 \geq \dots \geq x_p \quad \text{and} \quad y_1 \leq \dots \leq y_q.$$

4.2. The Parametrization. Let $p \times q$ denote the $p \times q$ grid, i.e. be the rectangular arrangement of squares with p rows and q columns.

Nested diagrams (λ, μ) . To an element $X \in \mathfrak{it}_0$, Bergeron [Ber09, §2] associates a pair (λ, μ) of Young diagrams by the rule

- The Young diagram $\lambda \subset p \times q$ corresponds to the sub-diagram of the $p \times q$ grid consisting of the boxes with coordinates (i, j) such that $x_i > y_j$.
- The Young diagram $\mu \subset p \times q$ corresponds to the sub-diagram of the $p \times q$ grid consisting of the boxes with coordinates (i, j) such that $x_i \geq y_j$.

Such a pair (λ, μ) will be called a *nested diagram*.

Example 4.1. With $p = 2, q = 3$ and $X = (3, 2, 1, 2, 3)$ the associated nested diagram (λ, μ) is given in the grid below.

λ, μ	λ, μ	μ
λ, μ	μ	

We also attach a coloring of the $p \times q$ diagram for each nested diagram (λ, μ) . The (i, j) -coordinate of the grid is colored in blue if it belongs to λ , white if it belongs to $\mu \setminus \lambda$ and red otherwise. With this convention, the coloring of the grid in example 4.1 is given by

Definition 4.2. Let $R(p, q)$ denote the set of all nested diagram (λ, μ) contained in the $p \times q$ grid. If (λ, μ) is the nested diagram associated to the element $X \in \mathfrak{it}_0$, then let R_X denote the colored $p \times q$ grid that corresponds to (λ, μ) .

Example 4.3. If $p = 3$ and $q = 4$ and $X = \{2, 2, 1, 1, 1, 2, 2\}$ then

$R_X =$

The main result of this paper is the following:

Theorem 4.4. *There is a bijective correspondence between $R(p, q)$ and $\mathcal{P}_c(p, q)$.*

Combined with Lemma 3.4, this gives the promised correspondance between the two parametrizations.

4.3. Bergeron's Lemma. As a first step, we give a proof of a result implicit in Bergeron's work, which characterizes permissible representations without reference of $\mathfrak{it}_0$.

Lemma 4.5. *The nested diagram (λ, μ) associated to an element $X = (x_1, \dots, x_p, y_1, \dots, y_q) \in \mathfrak{it}_0$ with $x_1 \geq \dots \geq x_p$ and $y_1 \leq \dots \leq y_q$ satisfies:*

- *The inclusions $\lambda \subset \mu \subset p \times q$*
- *The skew diagram $\mu \setminus \lambda$ is a union of rectangular diagrams $p_i \times q_i$, $i = 1, 2, \dots, m$ touching only at the corners.*

Conversely, given a pair of Young diagrams (λ, μ) satisfying the above, one can always find an element $X \in \mathfrak{it}_0$ such that (λ, μ) is the nested diagram associated to X .

Proof. Let X be as in the proposition and let (λ, μ) be the nested diagram associated to X . By definition $\lambda \subset \mu \subset p \times q$. Next, notice that a box with coordinates (i, j) is in $\mu \setminus \lambda$ if and only if $x_i = y_j$. To prove that $\mu \setminus \lambda$ consists of rectangles meeting at the corners, it suffices to prove the rectangles in $\mu \setminus \lambda$ cannot share a side.

Suppose $\mu \setminus \lambda$ has two rectangles which touch each other along one edge. Then there exists $2 \leq i \leq p$ and $1 \leq j \leq q-1$ such that $\mu \setminus \lambda$ contains the boxes with coordinates $(i, j-1), (i, j), (i-1, j)$ and does not contain the box with coordinates $(i-1, j-1)$. In this case,

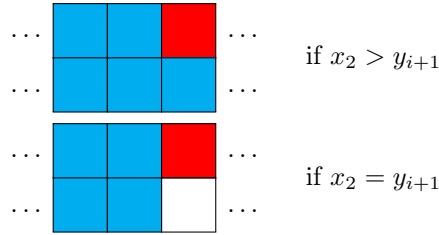
$$x_i = y_j = x_{i-1} > y_{j-1} = x_i \implies x_i > x_i$$

which is a contradiction. Therefore, $\mu \setminus \lambda$ cannot contain such rectangles. This establishes the second property of nested diagrams.

Conversely, suppose (λ, μ) is a pair of Young diagrams satisfying the requirements in the statement of the lemma, then it suffices to obtain from these Young diagrams a single string of inequalities consisting of x_i, y_j where $x_1 \geq \dots \geq x_p$ and $y_1 \leq \dots \leq y_q$ and then put $X = (x_1, \dots, x_p, y_1, \dots, y_q)$. To achieve this, we first color the $p \times q$ diagram using our method outlined above and assume that the variables y_i satisfy $y_1 \leq y_2 \leq \dots \leq y_q$. Then, using the coloring of the first row we can insert x_1 into the this inequality. We consider three cases.

Case 1. $y_{i+1} > x_1 > y_i$ for some $i \in \{1, \dots, q-1\}$.

In this case, we can read the second row and insert x_2 somewhere into this new string of inequalities. If it happens so that we obtain $x_2 \geq y_{i+1} > x_1 > y_i$ then the $p \times q$ grid must contain a subgrid of the form

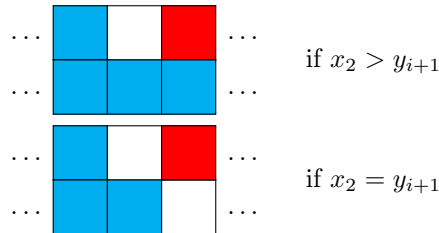


Since the blue boxes form the Young diagram λ , the obtained coloring when $x_2 > y_{i+1}$ contradicts with the order of elements in the partition that corresponds to λ . That is, the second row of a Young diagram cannot contain more boxes than the number of that of the first row.

Similarly, notice that the complement of the red boxes in the $p \times q$ grid forms the Young diagram μ . When $x_2 = y_{i+1}$, we see that the coloring contradicts with the order relations in the Young diagram μ .

In conclusion, we see that we must have $x_1 \geq x_2$ in case 1.

Case 2. $y_{i+1} > x_1 = y_i$ for some $i \in \{1, \dots, q-1\}$. In this case, if $x_2 \geq y_{i+1}$, we have two possible subgrids that must occur in $p \times q$.



In both situations, the coloring obtained contradicts with the order relations in the Young diagram λ so we must have $x_1 \geq x_2$.

Case 3. $y_{i+1} = x_1 = y_i$ for some $i \in \{1, \dots, q-1\}$. In this case, if $x_2 > y_{i+1}$ then there must exist a subgrid of the form



Again, the number of blue boxes in the second row must be greater than that of in the first row, which contradicts with the order relations of the Young diagram λ . Hence, $x_1 \geq x_2$.

Reading off each row and inserting x_j to the previously obtained string of inequalities, we end up with the relations $x_1 \geq \dots \geq x_p$ and $y_1 \leq \dots \leq y_q$ which means this Young diagram pair is obtained from the element $X = (x_1, \dots, x_p, y_1, \dots, y_q)$, hence (λ, μ) forms a nested diagram. This proves the lemma. $\square$

5. TRANSLATION BETWEEN PARAMETERIZATIONS

In order to prove a correspondence between the two parameterizations, we define a map $\alpha : \mathcal{P}_c(p, q) \rightarrow R(p, q)$ as follows. Given a maximally compact bipartition $B = ((p_1, q_1), \dots, (p_r, q_r)) \in P_c(p, q)$, produce the matrix

$$X_B = (\underbrace{r, \dots, r}_{p_1}, \underbrace{r-1, \dots, r-1}_{p_2}, \dots, \underbrace{1, \dots, 1}_{p_r}, \underbrace{1, \dots, 1}_{q_r}, \dots, \underbrace{r, \dots, r}_{q_1})$$

and set $\alpha(B) = R_{X_B}$. Theorem 4.4 will then follow from the following proposition:

Proposition 5.1. *The map $\alpha : P_c(p, q) \rightarrow R(p, q)$ is a bijection.*

Definition 5.2. An ordered tuple $(x_1, \dots, x_p, y_1, \dots, y_q)$ of $p+q$ elements is called *permissible* if it satisfies the following relations.

- $x_i, y_j \in \mathbb{R}$ for all $i = 1, \dots, p, j = 1, \dots, q$
-

$$\begin{aligned} x_1 &= \dots = x_{p_1} = y_q = \dots = y_{q-q_1+1} > \\ x_{p_1+1} &= \dots = x_{p_1+p_2} = y_{q-q_1} = \dots = y_{q-q_1-q_2+1} > \dots > \\ x_{p_1+\dots+p_{r-1}+1} &= \dots = x_{p_1+\dots+p_r} = y_{q_r} = \dots = y_1. \end{aligned}$$

Let $T(p, q)$ denote the set of all permissible $(p+q)$ -tuples.

Given a bipartition $B = ((p_1, q_1), \dots, (p_r, q_r)) \in \mathcal{P}(p, q)$, assign to it any element $a \in T(p, q)$. It is then easy to see that $\alpha(B) = R_{X_B} = R_X$ where $X = \text{diag}(a)$.

Example 5.3. To see the above algorithm in action, consider the bipartition $B = ((2, 1), (0, 1), (1, 2)) \in P_c(3, 4)$. Then we associate to B any ordered tuple $(x_1, x_2, x_3, y_1, y_2, y_3, y_4)$ satisfying the relations

$$x_1 = x_2 = y_4 > y_3 > x_3 = y_2 = y_1$$

from which we get

$$\alpha(B) = \begin{array}{|c|c|c|c|} \hline \text{blue} & \text{blue} & \text{blue} & \text{white} \\ \hline \text{blue} & \text{blue} & \text{blue} & \text{white} \\ \hline \text{white} & \text{white} & \text{red} & \text{red} \\ \hline \end{array}$$

Proposition 5.4. α is injective.

Proof. Let $B, B' \in \mathcal{P}_c(p, q)$ such that $B \neq B'$, and write

$$B = ((p_1, q_1), \dots, (p_r, q_r)), \quad B' = ((p'_1, q'_1), \dots, (p'_s, q'_s))$$

Assign a tuple $a = (x_1, \dots, x_p, y_1, \dots, y_q) \in T(p, q)$ to B and another tuple $b = (x'_1, \dots, x'_p, y'_1, \dots, y'_q) \in T(p, q)$ to B' . Without loss of generality, we may assume that $(p_1, q_1) \neq (p'_1, q'_1)$. We consider the following cases

- **Case 1:** Suppose $p_1 = p'_1$, so that without loss of generality $q_1 < q'_1$. If $p_1 = p'_1 = 0$, then B, B' being elements of $P_c(p, q)$ implies that $p_2, p'_2 > 0$. In particular, we have that

$$\begin{aligned} y_q &= \dots = y_{q-q_1+1} > x_1 = \dots \\ y'_q &= \dots = y'_{q-q'_1} = \dots = y'_{q-q'_1+1} > x'_1 = \dots \end{aligned}$$

Note that by the first string of inequalities we necessarily have $x_1 \geq y_{q-q_1}$, so the $(1, q-q_1)$ box in $\alpha(B)$ is either blue or white. On the other hand, $y'_{q-q'_1} > x'_1$, so the $(1, q-q'_1)$ box in $\alpha(B')$ is red. In either case, $\alpha(B) \neq \alpha(B')$. The case where $p_1 = p'_1 > 0$ follows by an identical argument,

where instead one shows that $x_{p_1+1} \geq y_{q-q_1+1}$ while $y'_{q-q_1+1} > x'_{p_1+1}$, so that $\alpha(B)$ and $\alpha(B')$ differ in the coloring of the box with coordinates $(p_1 + 1, q - q_1)$.

- **Case 2:** Suppose $p_1 \neq p'_1$, and without loss of generality $p_1 < p'_1$. If $p_1 = 0$, then $p_2 > 0$ since $B \in \mathcal{P}_c(p, q)$. The inequalities associated to a and b thus appear as

$$y_q = \cdots = y_{q-q_1+1} > x_1 = \cdots$$

$$x'_1 = \cdots = x'_{p_1} = y'_q = \cdots = y'_{q-q'_1+1} > \cdots$$

The first string of inequalities tells us that $y_q > x_1$, so the box in the $(1, q)$ coordinate in $\alpha(B)$ is colored in red. On the other hand, the second string of inequalities tells us that $x'_1 = y'_q$, so the box in the $(1, q)$ coordinate of $\alpha(B')$ is colored in white. Hence $\alpha(B) \neq \alpha(B')$. The case where $p_1 > 0$ follows from an identical argument where we obtain $y_q > x_{p_1+1}$ and $y'_q = x'_{p_1+1}$. This shows that the box in the $(p_1 + 1, q)$ coordinate of $\alpha(B)$ is colored in red and the box in the same coordinate of $\alpha(B')$ is colored in white, hence $\alpha(B) \neq \alpha(B')$.

This proves that if $B \neq B'$ then $\alpha(B) \neq \alpha(B')$, hence α is injective. $\square$

Proposition 5.5. *Each nested diagram (λ, μ) can be represented as a single string of inequalities, consisting of $p + q$ variables $x_1, \dots, x_p, y_1, \dots, y_q$.*

Proof. Let (λ, μ) be a nested diagram. Then by construction (λ, μ) is associated to an element $X = (x_1, \dots, x_p, y_1, \dots, y_q) \in \mathfrak{it}_0$. Our goal is to find a single string of relations satisfied by all elements $X' \in \mathfrak{it}_0$ whose associated nested diagram is (λ, μ) .

Firstly, we require the following inequalities

$$x_1 \geq \cdots \geq x_p \quad \text{and} \quad y_1 \leq \cdots \leq y_q$$

Next, each box in the grid gives a relation between two elements x_i, y_j . Using the first row, we insert x_1 into the string of inequalities $y_q \geq \cdots \geq y_1$. Since $x_2 \leq x_1$, using the second row, we insert x_2 into the previous chain of inequalities. The process continues until we reach the bottom row. At the end of this process, we obtain a string of inequalities that has $=, \geq$ and $>$ signs. Finally, we replace each $\geq$ sign with $=$ to get the desired string that will, in the next proposition, produce the corresponding bipartition for this particular nested diagram (λ, μ) . $\square$

Example 5.6. Consider the following coloring of the 3×4 grid.

By 4.5, this colored grid corresponds to a certain nested diagram (λ, μ) , which is associated to a certain element $X = (x_1, \dots, x_p, y_1, \dots, y_q) \in \mathfrak{it}_0$. The coloring of the grid helps us to read the following relations between the diagonal entries of X implied by the first two rows.

$$x_1 = y_4 > y_3 \geq y_2 \geq y_1$$

$$y_4 > x_2 = y_3 > y_2 \geq y_1$$

Applying the algorithm explained in the above proof then allows us to obtain the following relation among all diagonal elements of X .

$$y_4 \geq y_3 \geq y_2 \geq y_1$$

$$y_4 = x_1 > y_3 \geq y_2 \geq y_1$$

$$y_4 = x_1 > x_2 = y_3 > y_2 \geq y_1$$

$$y_4 = x_1 > x_3 = x_2 = y_3 > y_2 \geq y_1$$

Finally, we change all the $\geq$ signs into $=$ to obtain

$$y_4 = x_1 > x_3 = x_2 = y_3 > y_2 = y_1$$

as the single line of inequalities that represents this nested diagram (λ, μ) . In particular, if the above relations are satisfied by the diagonal entries of an element $X = (x_1, \dots, x_p, y_1, \dots, y_q) \in \mathfrak{it}_0$ then the associated nested diagram is exactly (λ, μ) .

Proposition 5.7. *The map α is surjective.*

Proof. Let (λ, μ) be a nested diagram. By the previous lemma, the corresponding $p \times q$ grid yields a single string of inequalities. Let $X = \{x_1, \dots, x_p\}, Y = \{y_1, \dots, y_q\}$. To find the corresponding bipartition that maps to (λ, μ) under α , we start by reading the string of inequalities from left to right and in the process forming tuples in the following manner:

- Let a_i (b_i) be the number of elements in X (correspondingly Y) that appear in between the $(i-1)$ -th and i -th $>$ signs. For $i = 1$, let a_1, b_1 be the number of elements that appear in X, Y before the first $>$ sign.

Then the bipartition that maps to the pair (λ, μ) is given by

$$B = \{(a_1, b_1), \dots, (a_r, b_r)\}$$

Now, we show that B is in fact a maximally compact bipartition. For the sake of contradiction, assume that the pairs $(1, 0), (1, 0)$ appear consecutively in B . Let's consider the single line of inequalities one obtains from B by using the method explained in the proof of Proposition 5.5. By the way B is constructed, there must exist some $1 \leq j \leq p-1$ such that the string " $\dots > x_j > x_{j+1} > \dots$ " appears in the single line of inequalities one constructs from (λ, μ) . Firstly, since the second input of $(1, 0)$ is 0, there are no $i \in \{1, \dots, q\}$ with $y_i = x_j$ or $y_i = x_{j+1}$. Secondly, since $(1, 0), (1, 0)$ appear consecutively in B , there is no $i \in \{1, \dots, q\}$ with $x_j > y_i > x_{j+1}$. Hence, such an occurrence of two consecutive $(1, 0)$ pairs must be obtained from string of inequalities that contains the subset $y_i > x_j > x_{j+1} > y_{i-1}$. However, by the algorithm explained in the proof of Proposition 5.5, each $\geq$ sign is turned into a $>$ sign in the end. Therefore, we must have had $x_j \geq x_{j+1}$ initially before making the change on the inequality signs. But the algorithm turns $\geq$ into $=$ hence B cannot contain such a subset of inequalities. Hence, $(1, 0), (1, 0)$ and by a similar reasoning $(0, 1), (0, 1)$ cannot occur in the bipartition B . A similar argument shows that pairs of the form $(m, 0), (n, 0)$ or $(0, m), (0, n)$ do not appear in B . Hence, B is a maximally compact bipartition.

This proves α is surjective. □

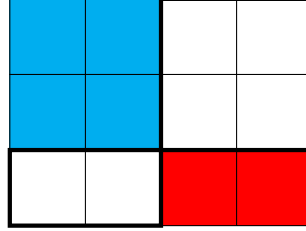
Propositions 5.4 and 5.7 prove that α is a bijective map from $P_c(p, q)$ to $R(p, q)$.

Example 5.8. For the grid given in example 4.3, the corresponding string of inequalities is $y_4 = y_3 = x_1 = x_2 > y_2 = y_1 = x_3$ and therefore the corresponding bipartition is $((2, 2), (1, 2)) \in P_c(3, 4)$. For example 5.6, the corresponding bipartition is given by $((1, 1), (2, 1), (0, 2)) \in P_c(3, 4)$.

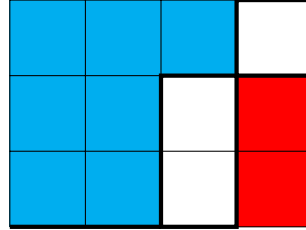
6. FAST WAY TO READ THE BIPARTITION FROM THE DIAGRAM

The above map provides a bijection from $\mathcal{P}_c(p, q)$ to $R(p, q)$. In the final section, we introduce a fast way to read off the bipartition corresponding to a given nested diagram (λ, μ) . Given the coloring of the $p \times q$ grid corresponding to a nested diagram (λ, μ) , one can obtain corresponding the bipartition by reading the sizes of the white-colored boxes from top-right to bottom-left. During this process, tuples of the form $(0, a)$ correspond to a white rectangle of size $0 \times a$, i.e. moving to left by a units without encountering any white boxes. Similarly, tuples of the form $(b, 0)$ correspond to white rectangles of size $b \times 0$, i.e. moving down by b units without encountering any white boxes. The reason behind this is that given a coloring of the $p \times q$ grid that corresponds to a nested diagram (λ, μ) , the white boxes denote the places that have $=$ in the single string of inequalities obtained from this grid using the method explained in 5.5. The number of vertical string of white boxes indicate the number of the x_i that are equal to each other, whereas the number of horizontal string of white boxes indicate the number of the y_j that are equal to each other. This gives the same pair (a_i, b_i) constructed in the proof of 5.7.

Example 6.1. The method above applied to the colored grid of example 4.3 gives us the bipartition $((2, 2), (1, 2))$



The colored grid of example 5.6 corresponds to the bipartition $((1, 1), (2, 1), (0, 2))$ as can be seen by tracing the thick borders in the below grid.



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